PakRoad-FloodWatch: Executive Project & Team Guide

Event: HackInsight 2026: Data-to-Decision (72-Hour Challenge) | **Track:** Climate (Urban Waterlogging & Mobility)

> **URGENT GATE:** The **Dataset Declaration** must be committed to your GitHub repository before **Monday, 19 October 2026 at 12:00 PM PKT**.

1. What Is Our Project? (The 1-Minute Story)

During monsoons in Pakistani cities (Lahore, Karachi, Rawalpindi), heavy downpours overwhelm drainage systems, submerging underpasses and arterial highways under 2–4 feet of water. Municipal authorities and traffic wardens typically react only *after* vehicles are already stranded and roads are paralyzed.

PakRoad-FloodWatch is an AI-driven early warning and decision-support system for **City Traffic Police** and **Rescue 1122**. It forecasts rainfall 1 to 24 hours ahead, integrates antecedent soil saturation and river discharge, and computes a 4-tier **Road Inundation Risk Index**. This allows wardens to place road-diversion cones and pre-deploy dewatering suction pumps **2 to 4 hours before gridlock occurs**.

2. Current Dataset Status (100% Collected & Validated)

We have collected and compiled a massive multi-source warehouse covering all 4 Pakistani provinces, Islamabad, and Gilgit-Baltistan:

Data Dimension	Coverage & Sources	Temporal Span	Volume	Status
Meteorological &	12 Major	2021-01-01 to	**592,176**	Cleaned

Data Dimension	Coverage & Sources	Temporal Span	Volume	Status
Soil Moisture	Cities (Lahore, Karachi, Rawalpindi, Islamabad, Faisalabad, Multan, Peshawar, Quetta, Hyderabad, Sukkur, Gilgit, Gwadar) via Open-Meteo	2026-10-18 (6 years, including 2022 super floods)	**hourly records**	& Indexed in SQLite
Hydrological Streamflow	5 Major River Basins (Ravi,	2021-01-01 to 2026-10-18	**10,585 daily discharg**	Cleaned & Indexed

Data Dimension	Coverage & Sources	Temporal Span	Volume	Status
	Indus, Kabul, Soan, Malir) via Copernicus / GloFAS		**e readings**	in SQLite
Spatial Risk & Road Infrastructure	10 Datasets (Rural Access Index RAI, flood exposure RP10/50/100, hospital buffers, evacuation capacity) via	All 160 Administrative Districts of Pakistan	**160 unified district profiles**	Joined on ADM2 PCODE

Data Dimension	Coverage & Sources	Temporal Span	Volume	Status
	UN OCHA / HDX			

> **Key Architecture Milestone:** The entire warehouse is compiled in `data/hackinsight.db` and tracked with SHA-256 hashes in `manifest.csv`. Running **`make data`** rebuilds the entire database from scratch idempotently in under 15 seconds.

3. The 4 Road Condition Severity Levels We Output

Level	Condition	Historical %	Operational Action Required
Level 0	**Normal / Dry**	95.23% (563,958 hrs)	Standard traffic monitoring. Roads clear and safe.
Level 1	**Wet / Caution**	4.35% (25,749 hrs)	Slippery road warnings issued to commuters; reduce speed limits.

Level	Condition	Historical %	Operational Action Required
Level 2	**Waterlogging Risk**	0.35% (2,093 hrs)	Alert WASA / Municipal pump teams to standby; monitor drainage bottlenecks.
Level 3	**Severe Inundation / Impassable**	0.06% (376 hrs)	Immediate road diversion; close underpasses; deploy heavy suction pumps.

4. Team Division (4 Members)

Member	Role	Key Responsibility
Member A	Domain & RAG Specialist	Curate NDMA / Rescue 1122 contingency PDFs; build the Insight Agent (RAG); implement Urdu (اردو) output for +2 bonus points.
Member B	Data Warehouse Engineer	Maintain `make data`, verify SHA-256 checksums in `manifest.csv`, manage SQLite schema in `data/hackinsight.db`.
Mem	Agent & Automation	Implement Data Quality Agent (Fix/Fill/Flag decision matrix); set up Prefect hourly

Member	Role	Key Responsibility
Member C	Lead	pipeline for 24-hr evidence window.
Member D	Model & ML Engineer	Train predictive model (LightGBM/XGBoost) using time splits; show uncertainty intervals; prove >=20% gain over declared baseline.

5. Declared Baseline (Locked In)

To adhere to Chapter 8 of the guidelines, our baseline is declared upfront before training:

> **Persistence Baseline:** $\hat{y}_t = y_{t-24}$
> *"Predicted rainfall at hour t equals the observed rainfall at the same hour yesterday ($t-24$)."*
> **Success Criterion:** Our ML model must beat the baseline MAE and Brier score by at least **20%** on an untouched future test set (2025–2026).

6. Submission Timeline & Key Milestones

- **Mon 19 Oct, 12:00 PM PKT:** Dataset Declaration committed to GitHub repository.
- **Tue 20 Oct, 4:00 PM to Wed 21 Oct, 4:00 PM PKT:** Mandatory 24-Hour continuous Prefect pipeline run window.
- **Wed 21 Oct, 11:59 PM PKT:** Final code commit tagged as `v1.0-final`.
- **Thu 22 Oct, 12:00 AM PKT:** Submission portal closes (Repo URL, HF Space link, 1-page Summary.pdf, 3-min video, AI_Use_Log.md).

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